

# RF Sensing and Holography in Dense Wireless Networks: Towards Privacy-Scalable and Ethical Design

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## Abstract

Recently, the development of techniques to capture and process wireless stray electromagnetic (EM) radiation from different radio sources is gaining increasing attention. RF sensing is, for example, an emerging paradigm that transforms existing wireless networks by adding sensing modalities to improve the perception of users and the environment. Stray fields of wireless devices in arbitrary frequency bands are recorded in a phase-sensitive manner and visualized in 3D to reconstruct or map the surrounding environment. These techniques can be generally exploited to transform radio networks into virtual radio sensors to allow mapping of objects, human-scale sensing, behavior recognition, and crowd density estimation. Ubiquitous perception through RF signals is a pivotal opportunity for future technology: it enables personalized services such as smart living, automated logistics or interaction through free-space gestures. However, it also challenges ethical and moral boundaries and threatens privacy. The workshop encourages authors from academia and industry to submit manuscripts on innovations on radio sensing, holography, networks, and computing techniques for privacy-selective human scale sensing. Analysis of social implications of the technology are also welcomed.

## CCS Concepts

• **Human-centered computing;**

## Keywords

RF sensing, EM modelling, holography, industry 5.0, human-machine interfaces, machine learning, wireless networks, Internet of Things, public awareness, ethical and social interaction studies

## ACM Reference Format:

Stefano Savazzi, Dariush Salami, Alexander Helmut Paulus, Luca Possati, and Riccardo Bersan. 2025. RF Sensing and Holography in Dense Wireless Networks: Towards Privacy-Scalable and Ethical Design. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous*

*Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3714394.3750582>

## 1 Objective of the workshop

Next generation industry, driven by emerging information technologies, will rely on advanced sensing tools as well as networked machines with increasing levels of intelligence and autonomy. Radio-Frequency (RF) sensing is an emerging paradigm that transforms existing industrial wireless networks by adding sensing modalities to improve the perception of workers, monitor the specific industrial workflow, and the surrounding workspace. Radio signals encode a view of all moving/fixed objects traversed by their propagation, and such view can be recovered at the receiver(s) side by imaging and data analytic tools [1, 2]. Through holographic, tomographic, and radar-based techniques radio signals are recorded in a phase-sensitive manner and visualized in 2D/3D to reconstruct or map the surrounding environment. Besides, new research opportunities will target the integration of RF sensing with distributed learning tool, and robust machine-level decision operations. Distributed Learning (DL) techniques, namely decentralized training (DT), federated or split learning (FL), are becoming crucial to enable challenging tasks in several workspace environments, improving safety, trustworthiness, privacy and data-ownership protection [3].

The combination of advanced RF sensing and distributed processing techniques can be generally exploited in industrial or smart home scenarios to allow mapping of objects, human-robot/human-machine collaborative situations, interaction through free-space gestures, activity or behavior recognition. Potential misuse or possible function creeping which challenges ethical and moral boundaries should be considered.

The workshop aims to put together original research papers on advanced algorithms for ubiquitous sensing and distributed learning in both industrial, private and public environments, and performance optimization.

## 2 Topics

The workshop will cover the following topics, among others:



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ACM ISBN 979-8-4007-1477-1/2025/10  
<https://doi.org/10.1145/3714394.3750582>

- Radio-based vision, classical and synthetic aperture radar methods [4–6], image processing of RF wave-fronts, statistical or Bayesian processing for environment mapping and localization;
- Human-robot advanced interfaces, free-space user interfaces and human-machine interaction modelling [7];
- Electromagnetics (EM), physics-driven body models and propagation prediction for radio sensing, imaging and holographic methods [8] (e.g. simulation, augmentation);
- Advanced smart antenna supporting beamforming, beam-steering and reconfigurable surface systems for radio sensing;
- Sustainable, privacy and ethically compliant RF sensing based on Federated Learning tools;
- Novel radio sensing in mm-Wave, sub-THz (i.e., 75 - 260 GHz) bands: multi-spectral and multi-domain data-fusion techniques to improve the robustness of radio sensing systems;
- Combination, transformation, and hybridization of multiple RF sensing and networking technologies for location-based services, motion detection, indoor navigation, mapping and charting in industrial workspaces;
- Distributed machine learning and AI tools for computationally feasible radio sensing and monitoring of industrial sites, including federated learning methods in robotized environments and automated buildings;
- Applications of radio sensing in workplaces, smart living and Industry 5.0 scenarios, 3D mapping, free-space user interfaces and human-machine interactions [9–11];
- Public awareness and social interaction studies, considering privacy threats and ethical compliant designs (selective perception);
- RF sensing and possible biases in data representations and decision-making, impact of dataset limitations and demographic exclusions;
- Strategies for ensuring fairness and equitable access in RF sensing applications.

### 3 RF sensing in Industry 5.0

The ability of machines to perceive and understand the surrounding environment has been one of the enabling factors of modern automation. This ability has proven to help overcome the existing barrier and change the paradigms historically characterized automation, improving the production system efficiency and working conditions [12, 13]. In this field, technologies considered innovative a few years ago are essential in modern production systems today. The degree of maturity reached by these technologies allowed us to realize a critical analysis of them to identify not just the advantages but also the limits and the risks associated. The primary example could be represented by the machine vision, a technology that, despite constant evolution driven by artificial intelligence techniques, is beginning to show important limits essentially linked to the ethical use of the technologies, the strong dominance of the foreign market, and the poor performance in dirty or polluted environments. In wondering how to overcome these limits, the industrial and research world are looking today to new systems working upon

different physical principles. Among emerging technologies that are becoming popular in research and partially in the industrial sector, RF sensing shows important potential [14]. The approach to this technology is multidisciplinary, and the objectives sought in its use are multiple. The same system can be helpful not only for allowing a machine to perceive the surrounding environment but also for communication purposes. At the same time, this technology shows great pervasiveness; think of the enormous diffusion of WiFi antennas, which are not generally perceived as a threat and do not allow a single person to be directly identified (unless voluntarily). This type of system is also highly robust and particularly suitable for industrial environments where sometimes unclean places or present obstacles limit the use of vision or laser systems. Finally, the possibility of using the same system also for communication purposes (as WiFi) or to realize a natural language interface machines enhances the multidisciplinary capabilities of this technology.

The proposed workshop aims to collect different experiences in the field of RF-sensing to define the current and future role of this technology and immediately define its advantages and critical issues considering its current and future role targeting in industry and automation as promising areas.

### 4 Ethical design patterns in RF sensing and Holography

RF sensing solutions may soon reach comparable quality as classical vision approaches and even surpass vision-based techniques. Interpreting stray radiation from ubiquitous wireless devices might target the reconstruction of 3D environments to resolve objects or individuals inside crowds and under variable environmental conditions, i.e., dark or occluded spaces, and non-Line-of-Sight (NLOS). On the other hand, integrating RF holography into ubiquitous communication systems may have unforeseen social and ethical consequences, both positive and negative. To limit possible negative outcomes, novel research directions will consider design paradigms that may constrain system capabilities, such as limiting the sensing range, obfuscation of unauthorized areas and enforcing the individual's right to be excluded (or be forgotten) by the sensing system.

The need for novel “ethically scalable” RF sensing tools is expected to emerge soon, since the technology is evolving and already integrated in various forms in research and patents. Future development of the technology therefore demands an effort to embed ethics into the design and application of what may soon be an enabling technology – a technology that provides the basis and support for other systems.

Beside exploring recent research and technological advancements in the RF sensing field, as described in the previous section, the workshop will also explore challenges and opportunities of ethics-by-design in RF sensing, examining design models, technical and ethics measures, their potential role in future regulatory efforts.

### 5 Organizers' backgrounds

- **Stefano Savazzi** is Senior Researcher at Consiglio Nazionale delle Ricerche (CNR), the Institute of Electronics, Computer and Telecommunication Engineering (IEIT). He received the M.Sc. degree and the Ph.D. degree (Hons.) in ICT from the

Politecnico di Milano, Italy, in 2004 and 2008, respectively, and joined CNR in 2012. He was a Visiting Researcher with Uppsala University, in 2005 and University of California at San Diego in 2007. He has coauthored over 110 scientific publications (Scopus). His current research interests include distributed signal processing, machine learning and networking aspects for the Internet of Things, radio localization and vision technologies. Dr. Savazzi was the recipient of the 2008 Dimitris N. Chorafas Foundation Award. He is CNR Principal Investigator in Horizon EIC Pathfinder project Holden, Horizon EU TRUSTroke and Horizon Marie Curie SmartTest projects. He is also serving as Associate Editor for Frontiers in Communications and Networks, Wireless Communications and Mobile Computing and Sensors.

- **Dariusz Salami** is a Radio Research Scientist at Nokia Bell Labs and a Postdoctoral Researcher at Aalto University, specializing in AI/ML for wireless communications, sensing, and 6G networks. He received his Ph.D. from Aalto University in 2023 as a Marie Skłodowska-Curie Fellow, focusing on AI/ML for wireless communication and sensing. His Ph.D. thesis was recognized with the prestigious Aalto Dissertation Award. He has published extensively in top-tier journals and conferences, contributing to areas such as reinforcement learning for adaptive sensing, joint radar-communication systems, and human-centric mmWave sensing. He has led multiple research projects, secured funding from esteemed organizations like the Nokia Foundation, Google, Finnish Foundation for Technology Promotion, and HPY Grant. With a strong academic and industrial background, he is committed to advancing ethical AI and next-generation wireless technologies.
- **Alexander Paulus** is currently a Visiting Researcher at the Politecnico di Milano, Milan, Italy. He received the M.Sc. and the Dr.-Ing. degree in electrical engineering and information technology from the Technical University of Munich, Munich, Germany, in 2015 and 2022. Since 2022, he has been working as a Senior Researcher at the Chair of High-Frequency Engineering, Department of Electrical Engineering, School of Computation, Information and Technology of the Technical University of Munich. He is recipient of the 2024 IEEE AP-S R.W.P. King Award. His research interests include inverse electromagnetic problems and imaging, computational electromagnetics and antenna measurement techniques.
- **Luca Possati** is an Assistant Professor at the University of Twente in the Netherlands, specializing in human-technology interaction. He also holds a senior researcher position with the international research program Ethics of Socially Disruptive Technologies (ESDiT). His research focuses on the intersection of philosophy of technology, postphenomenology, and the psychology of technology, exploring how humans engage with and are shaped by technological systems. His methodological approach integrates insights from cognitive sciences, neuropsychology, design studies, and actor-network theory, allowing for a multidisciplinary analysis of the complex relationships between individuals, society, and technology. He has published numerous papers and books

on phenomenology, and history of contemporary philosophy. He is the author of *The Algorithmic Unconscious. How Psychoanalysis Helps in Understanding AI* (Routledge, 2021).

- **Riccardo Bersan** is the Director of System and Software Engineering at Adant, a company based in Padova that develops and commercializes a novel class of advanced smart antenna systems. Riccardo holds a Master's degree in Electronic Engineering from the University of Padua, Italy, earned in 2013. Since joining Adant, Riccardo has held various roles in system testing, software, and systems engineering. He has supervised multiple thesis projects, co-authored scientific articles and patents, and contributed to research collaborations with international universities. Currently, Riccardo leads and manages a team of software and system engineers to develop and maintain software that enables smart antenna technology, ensuring alignment with the company's strategic goals and meeting high standards of quality and performance.

## 6 Workshop Duration and Format

Half day Workshop, based on solicited research papers. These papers must report significant and innovative research and development results that will have a long-term impact on the field of research, with the potential for implementation. The final manuscripts must comply with the formatting requirements for UBICOMP 2025, with a page limit that depends on the submission type (see below), and a presentation slot of 30 minutes maximum. The working language of the conference is English. Accepted, registered, and presented papers will be made available in the ACM Digital Library.

Text-based version of the CFP (Call for Papers) is attached.

## 7 Submission Details

We will accept two types of submission: long and short papers. A long paper should have a length of maximum 6 pages and short paper maximum 4 pages. Papers should be submitted using the UbiComp ISWC 2025 proceedings format, see the UbiComp website for more details: <https://ubicomp.org/ubicomp-iswc-2025/authors/formatting/>

All submitted papers will be reviewed and judged on originality, technical correctness, relevance, and quality of presentation. We explicitly invite submissions of papers that describe preliminary results or work-in-progress. The accepted papers will appear in the UbiComp supplemental proceedings and in the ACM Digital Library.

Successful submissions will have the potential to raise discussion, provide insights for other attendees, and illustrate open challenges and potential solutions.

At least one author of each accepted paper needs to register for the conference and the workshop itself. During the workshop, each paper will be presented briefly by one of the authors. In addition, there will be room for demonstrations as well as discussions.

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